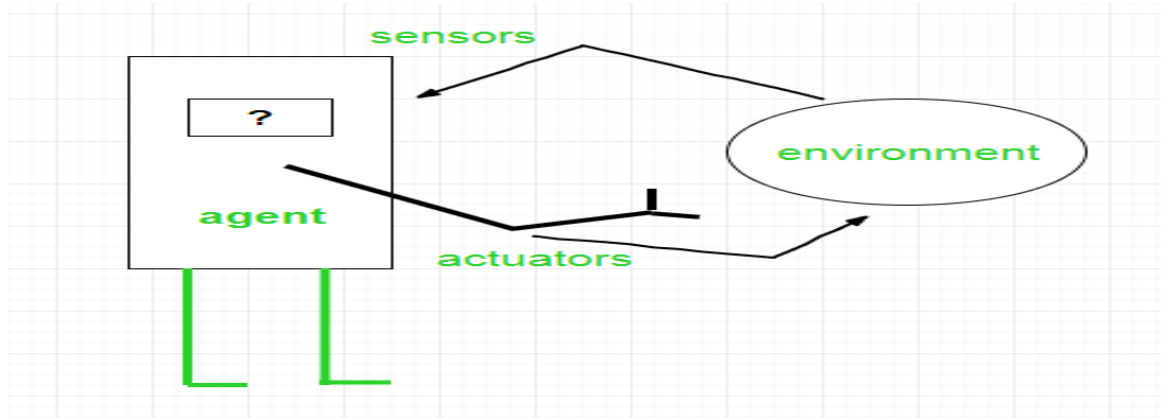


# THE STRUCTURE OF AGENTS



- **Architecture** is the **machinery** that the agent executes on. It is a device with sensors and actuators.
- **Agent program** is an implementation of an **agent function**.
- An **agent function** is a map from the percept sequence to an action.

**Agent = Architecture + Agent Program**

# Types of Agents

Agents can be grouped into five classes based on their degree of perceived intelligence and capability :

- Simple Reflex Agents
- Model-Based Reflex Agents
- Goal-Based Agents
- Utility-Based Agents
- Learning Agent

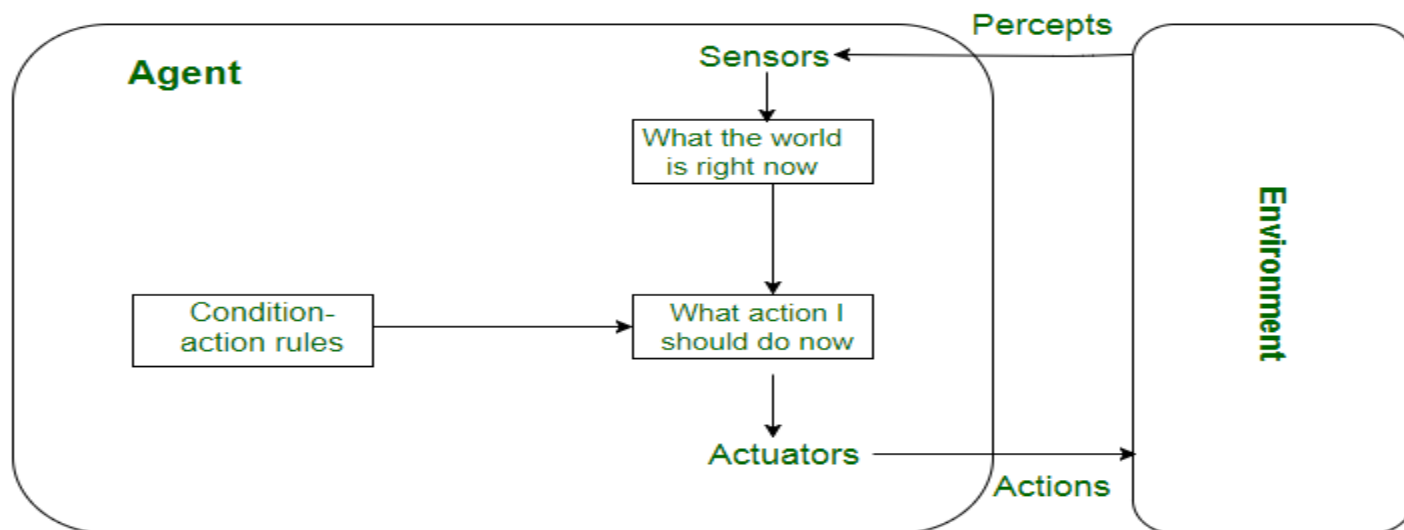
# Simple reflex agents

- Simple reflex agents **ignore the rest of the percept history** and **act** only on the basis of the **current percept**.
- **Percept history is the history of all** that an agent has perceived to date.
- The agent function is based on the **condition-action rule**. A condition-action rule is a rule that maps a state i.e, **condition to an action**.
- **If the condition is true, then the action is taken, else not**. This agent function only succeeds when the environment is fully observable.
- For simple reflex agents operating in **fully observable** environments, **infinite loops are often unavoidable**.

- It may be possible to escape from infinite loops if the agent can randomize its actions.

Problems with Simple reflex agents are :

- Very limited intelligence.
- No knowledge of non-perceptual parts of the state.
- Usually too big to generate and store.
- If there occurs any change in the environment, then the collection of rules need to be updated.



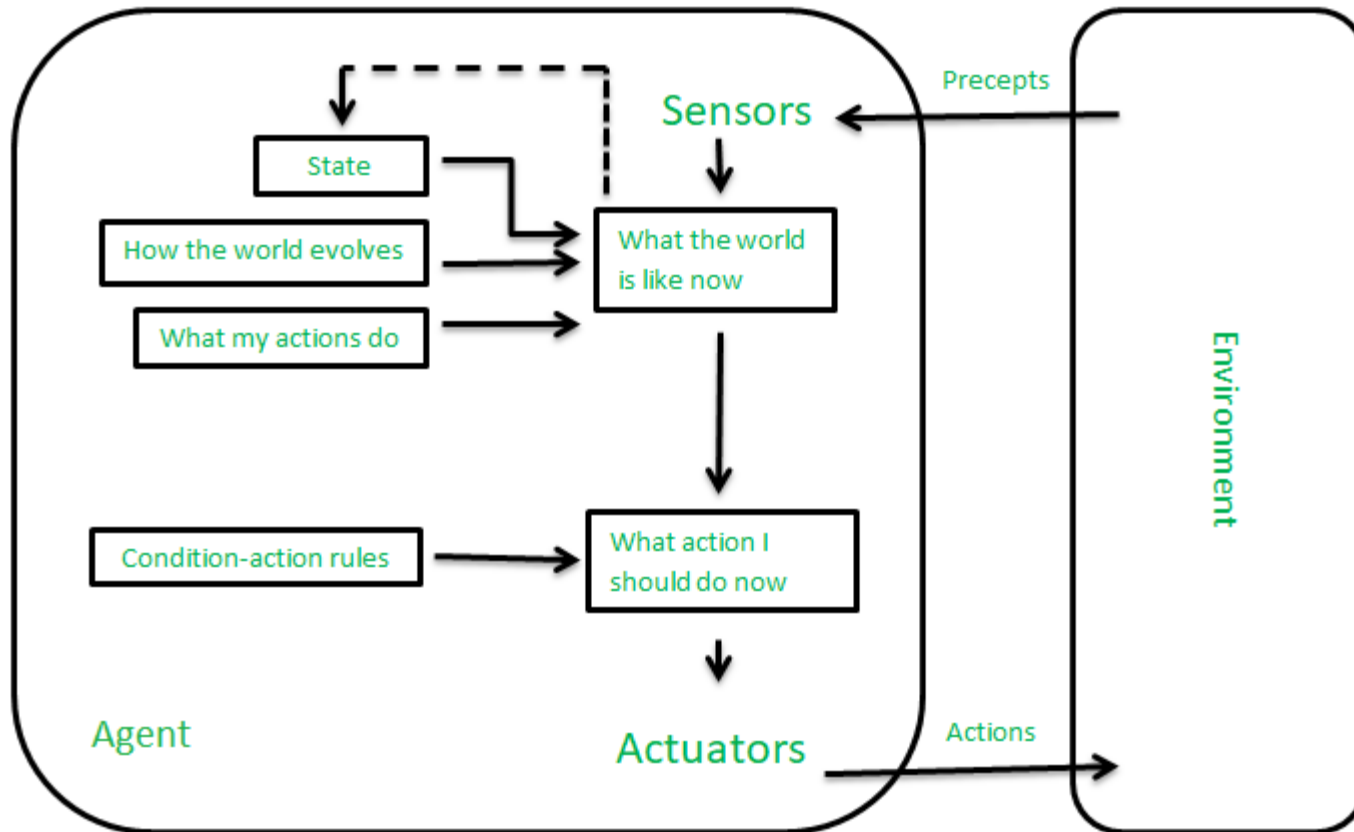
# Model-based reflex agents

- It works by finding a rule whose condition matches the current situation.
- A model-based agent can handle partially observable environments by the use of a model about the world.
- The agent has to keep track of the internal state which is adjusted by each percept and that depends on the percept history.
- The current state is stored inside the agent which maintains some kind of structure describing the part of the world which cannot be seen.

Updating the state requires information about :

- ❖ how the world evolves independently from the agent, and
- ❖ how the agent's actions affect the world.

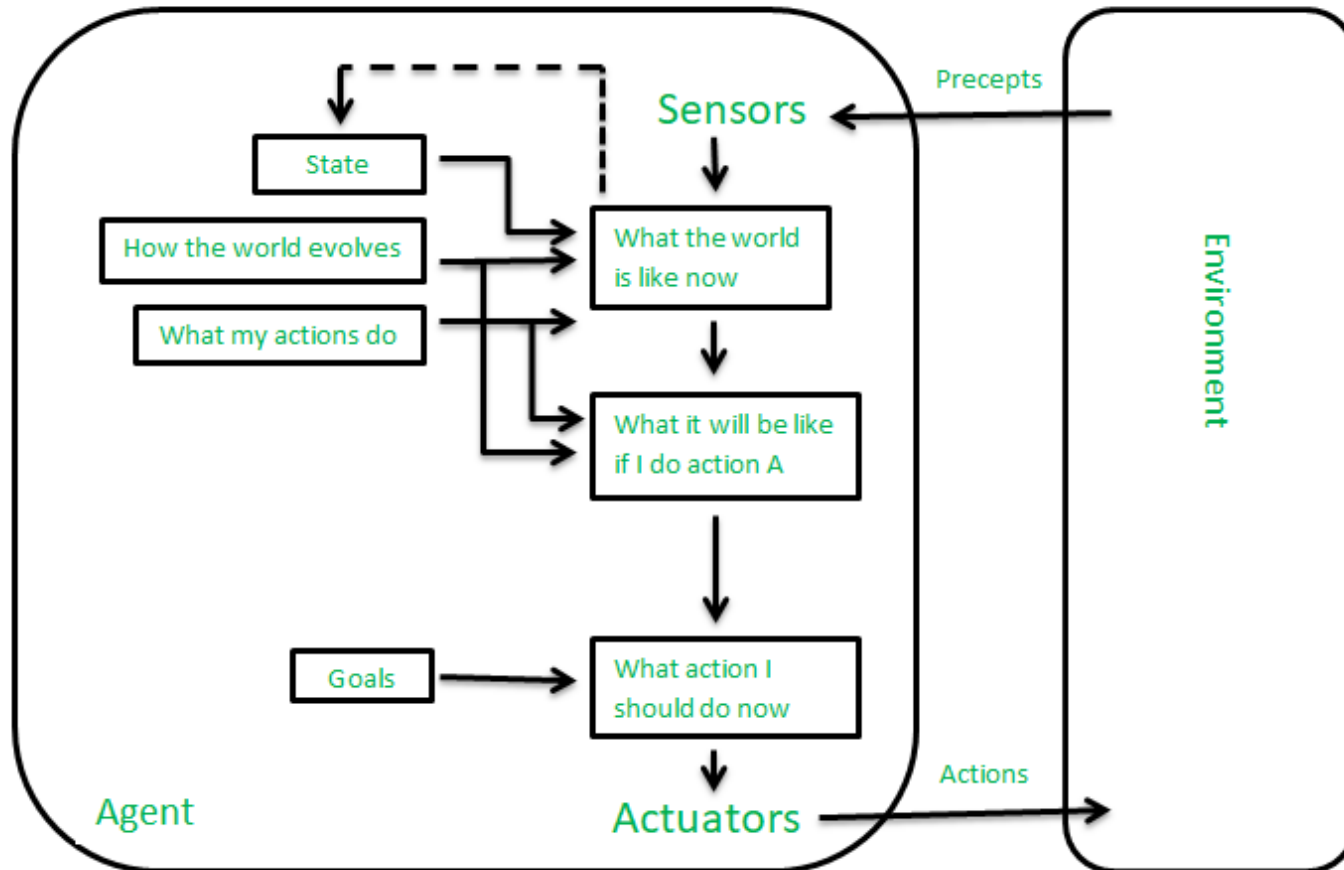
# Model-based reflex agents



# Goal-based agents

- These kinds of agents take decisions based on how far they are currently from their goal (description of desirable situations).
- Their every action is intended to reduce its distance from the goal.
- This allows the agent a way to choose among multiple possibilities, selecting the one which reaches a goal state.
- The knowledge that supports its decisions is represented explicitly and can be modified, which makes these agents more flexible. They usually require search and planning.
- The goal-based agent's behavior can easily be changed.

# Goal-based agents

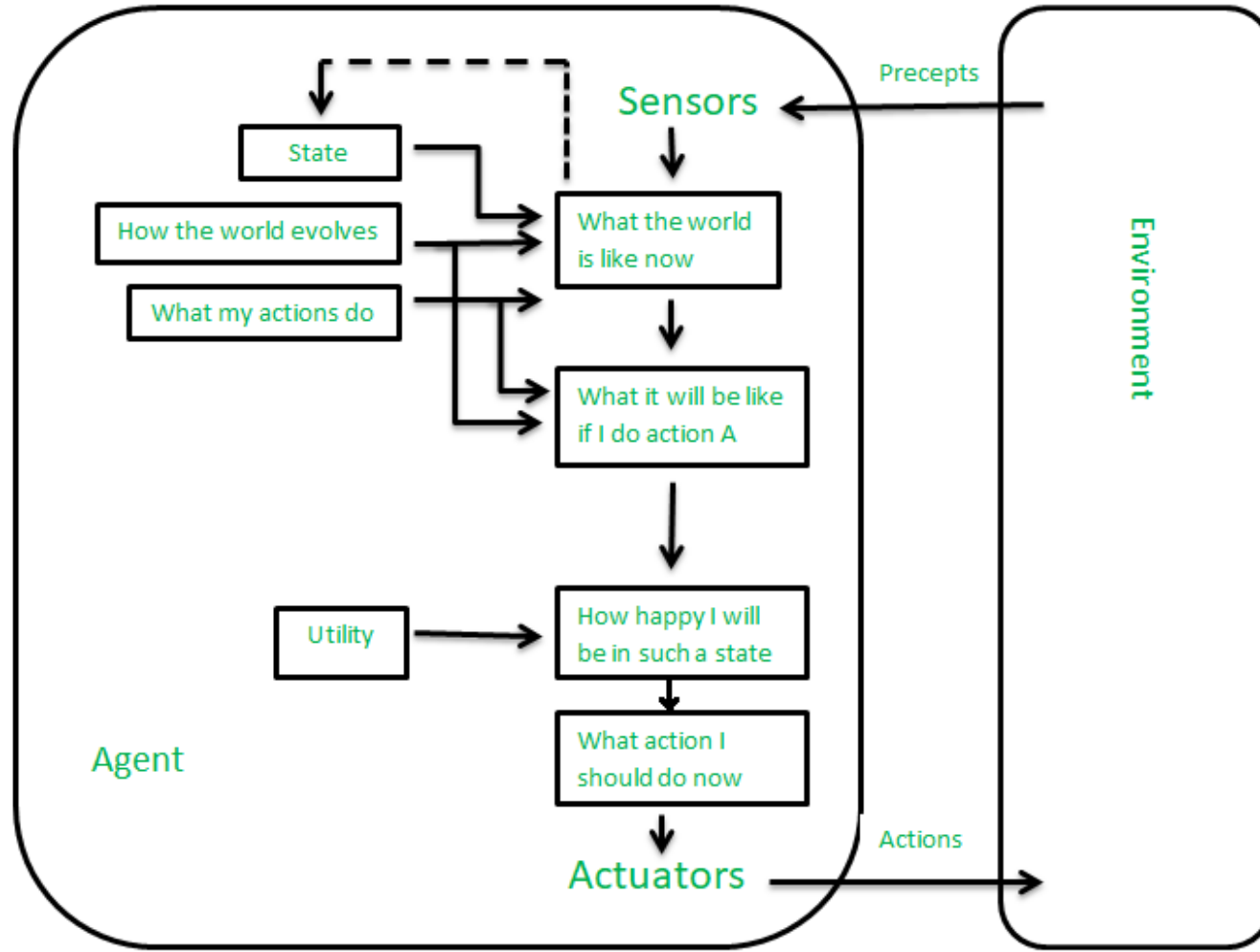




# Utility-based agents

- The agents which are developed having their end **users as building blocks** are called utility-based agents.
- When there are **multiple possible alternatives, then to decide which one is best**, utility-based agents are used.
- They choose actions based on a **preference (utility)** for each state.
- Sometimes **achieving the desired goal is not enough**. We may look for a **quicker, safer, cheaper trip to reach a destination**. Agent happiness should be taken into consideration.
- Utility describes how **“happy”** the agent is. Because of the uncertainty in the world, a utility agent **chooses the action that maximizes the expected utility**.
- A utility function **maps a state onto a real number which describes the associated degree of happiness**.

# Utility-based agents



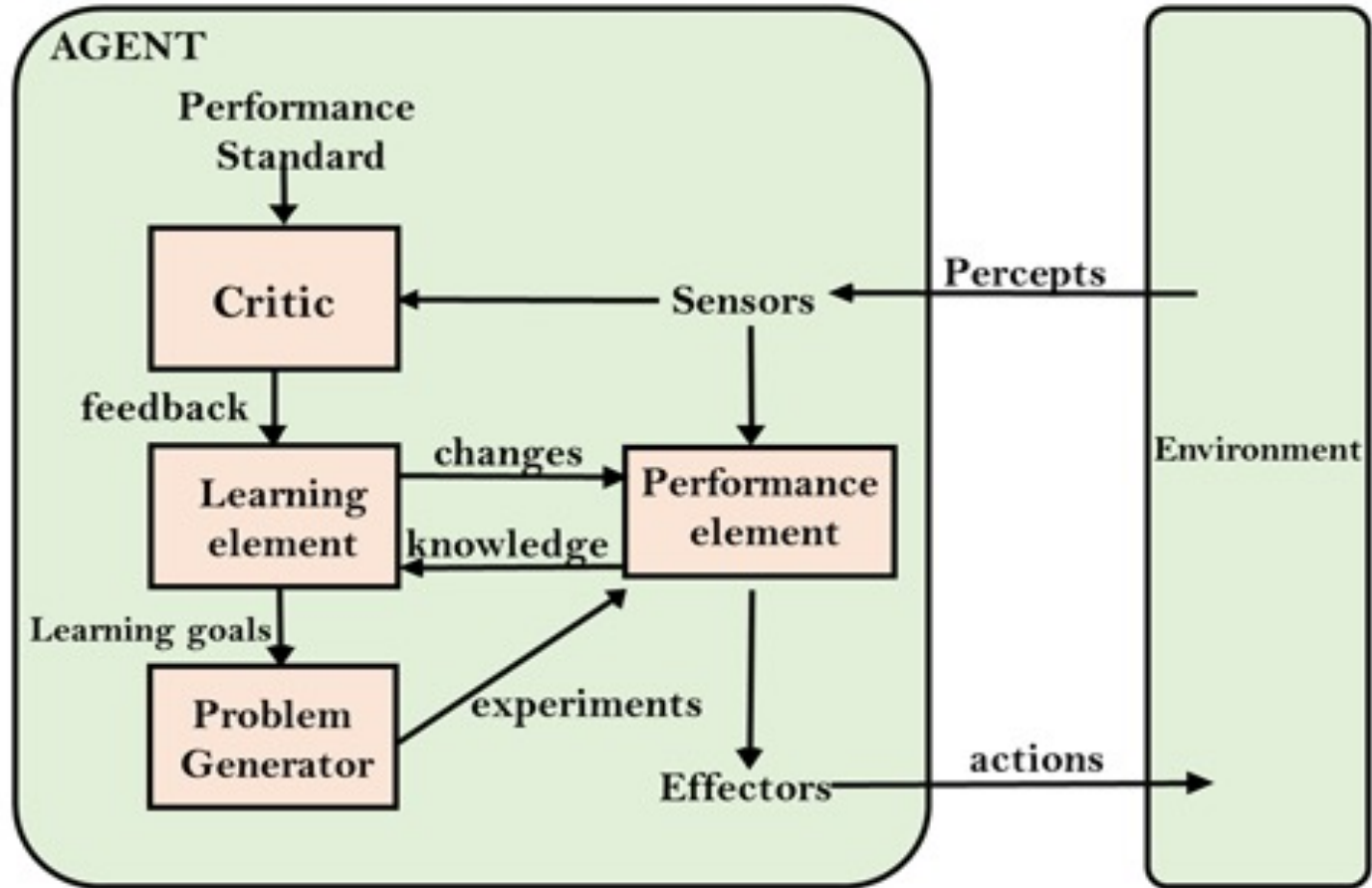
# Learning Agent

- A learning agent in AI is the type of agent that **can learn from its past experiences or it has learning capabilities**. It starts to act with basic knowledge and then is able to act and adapt automatically through learning.

A learning agent has mainly four conceptual components, which are:

- **Learning element:** It is responsible for making improvements by learning from the environment
- **Critic:** The learning element takes feedback from critics which describes how well the agent is doing with respect to a fixed performance standard.
- **Performance element:** It is responsible for selecting external action
- **Problem Generator:** This component is responsible for suggesting actions that will lead to new and informative experiences.

# Learning Agent



THANK YOU